



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 2 HOMEWORKS

SOLUTION - Q3

Version

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Instructor

Abiria

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Note 1 (Notations)

The following notations are used throughout the questions below.

- $\mathbb{Z}$: The set of all integers.
- $\mathbb{Z}^+$: The set of all positive integers; i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: The set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $\mathbb{R}$: The set of all real numbers.
- $\min(x, y)$: The minimum of x and y ; i.e., the smaller of the two.
- $\mathcal{P}(X)$: The power set of the set X .
- μ_n : The set of all n -th roots of unity.

Q1.

EASY

Let $f, g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ be given by

$$f(x) = x + 3, \quad g(x) = 2x, \quad h(x) = x - 1.$$

What is the value of $(h \circ g \circ f)(1)$?

- ☐ 1. 3
 ☐ 2. 5
 ☐ 3. 7
 ☐ 4. 8
 ☐ 5. 9

Q2.

EASY

Let S_5 be the set of all bijections from $\mathbb{Z}_5$ to $\mathbb{Z}_5$.

What is the value of $|S_5|$?

- ☐ 1. 5
 ☐ 2. 25
 ☐ 3. 120
 ☐ 4. 125
 ☐ 5. 625

Definition 2 (Injection)

A function $f : X \rightarrow Y$ is said to be *injective*, or called an *injection*, if for every $x, y \in X$ with $x \neq y$, we have $f(x) \neq f(y)$.

Q3.

EASY

Classify each of the following functions as injective or NOT injective. Which one differs from the rest?

- ☒ 1. $f_1 : \mathbb{R} \rightarrow \mathbb{R}, f_1(x) = \min(x, 2x)$
☐ 2. $f_2 : \mathbb{R} \rightarrow \mathbb{R}, f_2(x) = 0 \cdot x$
- ☐ 3. $f_3 : \mathbb{Z} \rightarrow \mathbb{Z}, f_3(x) = \gcd(x, 10)$
☐ 4. $f_4 : \mathbb{R} \rightarrow \mathbb{R}, f_4(x) = x^2 + 5x + 6$
- ☐ 5. $f_5 : \mathbb{Z} \rightarrow \mathbb{Z}, f_5(x) = |x|!$

- Note that $\min(x, 2x) = x$ for $x \geq 0$ and $\min(x, 2x) = 2x$ for $x \leq 0$, so f_1 is built from two strictly increasing lines that agree at $x = 0$. Hence f_1 is strictly increasing on all of $\mathbb{R}$, and a strictly increasing function never takes the same value twice. Thus, f_1 is injective.
- There exist two distinct real numbers 1 and 2 such that $0 \cdot 1 = 0 \cdot 2 = 0$. Thus, f_2 is not injective.
- There exist two distinct integers 4 and 8 such that $\gcd(4, 10) = \gcd(8, 10) = 2$. Thus, f_3 is not injective.
- Observe that the polynomial can be factored into $(x + 2)(x + 3)$. From here, it is clear that there are two distinct real numbers -2 and -3 such that $f_4(-2) = f_4(-3) = 0$. Thus, f_4 is not injective.
- There exist two distinct integers 0 and 1 such that $|0|! = |1|! = 1$. Thus, f_5 is not injective.

Therefore, the first choice is the correct answer.

Definition 3 (Equality of functions)

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if $f, g : X \rightarrow Y$ are functions with the same domain and codomain, then $f = g$ if and only if $f(x) = g(x)$ for every $x \in X$.

Q4.

EASY

Given that every function below has $\mathbb{Z}^+$ as both its domain and codomain, which one differs from the rest?

- | | |
|---|--|
| ○ 1. $f_1(x) = \gcd(x, x^2) + \gcd(x, x^3)$ | ○ 2. $f_2(x) = \text{lcm}(2x, 3x) - \text{lcm}(x, 3x)$ |
| ○ 3. $f_3(x) = {}_{x+1}C_x + x - 1$ | ○ 4. $f_4(x) = \gcd(2x, 6x)$ |
| ○ 5. $f_5(x) = 3x - x $ | |

Definition 4 (Idempotent function)

A function f from a set X to itself is called *idempotent* when $f \circ f = f$.

Q5.

EASY

Which of the following functions is NOT idempotent?

- | | |
|--|--|
| ○ 1. $f_1 : \mathbb{Z} \rightarrow \mathbb{Z}, f_1(x) = x \bmod 5$ | ○ 2. $f_2 : \mathbb{R} \rightarrow \mathbb{R}, f_2(x) = \min(\sqrt{2}, x)$ |
| ○ 3. $f_3 : \mu_{64} \rightarrow \mu_{64}, f_3(x) = x^8$ | ○ 4. $f_4 : \mathbb{R} \rightarrow \mathbb{R}, f_4(x) = 2x - x $ |
| ○ 5. $f_5 : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R}), f_5(x) = x \cup \mathbb{Z}$ | |

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark $\times$ in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark $?$ in the Evaluation column.
- If you understood the question and are sure you have solved it, mark $\circ$ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q4	
Q2		Q5	
Q3			

There are also a few additional questions; feel free to answer them.

- (a) Which was the hardest for you, among those you have solved? _____
- (b) Which seemed to be the hardest for you, among those you haven't solved? _____
- (c) Which was your favorite question, if you had to choose one? _____